YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 3 SOLUTIONS

Proofs

SECTION ONE

1 a Converse: '*If Q then P*' Contrapositive: '*If not Q then not P*. **[2]**

b i *I am, therefore I think*. Answers may vary. **[1]**

ii *If I am not, then I don't think*. Valid if the original is valid. **[2]**

c i If n is not an even integer then n^2 is not an even integer. **[1]**

ii If n is not even, then it is odd.

Let $n = 2k + 1$ **[1]**

$\therefore \quad n^2 = (2k + 1)^2$

$= 4k^2 + 4k + 1$ **[1]**

$= 2(2k^2 + 2k) + 1$ **[1]**

$= 2m + 1$ where $m = 2k^2 + 2k$ which is an integer. **[1]**

$\therefore n^2$ is odd, not an even integer. **[1]**

2 i $-1 > -3$ but $(-1)^2 > (-3)^2$ is not true. **[1]**

ii 2 is a prime number that is not odd. **[1]**

iii If $p = 19$, then $p + 2 = 21$ which is not prime. Other examples possible. **[1]**

3 Prove: The sum of four consecutive odd integers is divisible by 8.

Any odd number can be written as $2n + 1$ where n is an integer. **[1]**

The next odd numbers will be $2(n + 1) + 1 = 2n + 3, 2n + 5, 2n + 7$. **[1]**

The sum will be $2n + 1 + 2n + 3 + 2n + 5 + 2n + 7 = 8n + 16$ **[1]**

$8n + 16 = 8(n + 2)$ which is divisible by 8 since $(n + 2)$ is an integer. **[2]**

Note: Proofs may vary.

4 a Any of the following.

All squares are rhombuses.

All rhombuses are parallelograms.

$\therefore$ All squares are parallelograms.

All rectangles are parallelograms

All parallelograms are quadrilaterals

$\therefore$ All rectangles are quadrilaterals

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All squares are rectangles

All rectangles are parallelograms

$\therefore$ All squares are parallelograms **[2]**

b Either of the two below.

All squares are parallelograms.

All parallelograms are quadrilaterals.

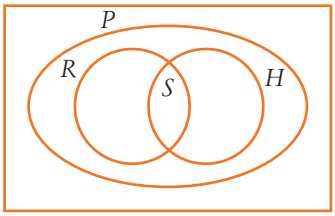
$\therefore$ All squares are quadrilaterals. **[2]**

All squares are rectangles.

All rectangles are quadrilaterals.

$\therefore$ All squares are quadrilaterals. **[2]**

c Q



[4]

5 a i The answer must be odd.

ii The answer must be even. **[2]**

b Let any two odd numbers be $2n + 1$ and $2m + 1$ and any two even numbers be $2n$ and $2m$, where n and m are integers. **[2]**

The product of two odd numbers is $(2n + 1)(2m + 1)$

$= 4mn + 2n + 2m + 1$

$= 2(2mn + n + m) + 1$ **[1]**

This is in the form $2k + 1$ where $k = 2mn + n + m$ is an integer, and so is odd. **[1]**

The product of two even numbers is $(2n)(2m) = 4mn = 2(2mn)$ **[1]**

This is in the form $2k$ where $k = 2mn$ is an integer, and so is even. **[1]**

c Suppose $\log_3^4$ is not irrational, i.e. is rational. **[1]**

Then $\log_3^4$ can be expressed as $\frac{a}{b}$ where a and b are integers. **[1]**

$\log_3^4 = \frac{a}{b} \Rightarrow 3^{\frac{a}{b}} = 4$ **[1]**

Raise both sides to the power b. **[1]**

$\left(3^{\frac{a}{b}}\right)^b = 4^b$ [1]

$3a = 4b$

The left side is a product of odd numbers and by part **i** must be odd. [1]

The right side is a product of even numbers and by part **ii** must be even.

[1]

An odd number cannot equal an even number.

So $\log_3^4$ cannot be rational and must therefore be irrational. [1]

6 Test for $n = 1$ [1]

L.S. $= 1 = 1^2 =$ R.S. so the conjecture is valid for $n = 1$. [1]

Assume the conjecture is valid for $n = k$.

i.e. $1 + 3 + 5 + 7 + \ldots + (2k - 1) = k^2$ [1]

Test for $n = k + 1$. [1]

Add $2(k + 1) - 1$ to both sides

$$\begin{aligned} 1 + 3 + 5 + 7 + \ldots + (2k - 1) + [2(k + 1)) - 1] &= k^2 + 2k + 2 - 1 \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$ [2]

$\therefore$ if the conjecture is valid for $n = k$, it is valid for $n = k + 1$. [1]

But, it is valid for $n = 1$ (so valid for $n = 2$, $n = 3$, etc.)

$\therefore$ it is valid for all natural numbers. [1]

7 a Test for $n = 1$.

R.S. $= a^1 - 1 = a - 1$ and $(a - 1)$ is a factor of $a - 1$. [2]

Assume the conjecture is valid for $n = k$.

i.e. $a - 1$ is a factor of $a^k - 1$.

This is equivalent to $a^k - 1 = m(a - 1)$, where m is an integer. [2]

Test for $n = k + 1$. [1]

$$\begin{aligned} a^{k+1} - 1 &= a^{k+1} - a + a - 1 \\ &= a(a^k - 1) + a - 1 \\ &= am(a - 1) + 1(a - 1) \quad \text{since } a^k - 1 = m(a - 1) \\ &= (am + 1)(a - 1) \end{aligned}$$ [3]

This is a multiple of $(a - 1)$ since $am + 1$ is an integer, hence $(a - 1)$ is a factor of $a^{k+1} - 1$.

$\therefore$ if the conjecture is valid for $n = k$, it is valid for $n = k + 1$. [1]

But, it is valid for $n = 1$ (so valid for $n = 2$, $n = 3$, etc.)

$\therefore$ it is valid for all natural numbers. [1]

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b i Assume the conjecture is valid for $n = k$.

i.e., $a + 1$ is a factor of $a^{2k+1} - 1$.

This is equivalent to $a^{2k+1} - 1 = m(a + 1)$, where m is an integer. **[2]**

Test for $n = k + 1$.

$$\begin{aligned} a^{2(k+1)+1} - 1 &= a^{2k+3} - 1 \\ &= a^{2k+3} - a^2 + a^2 - 1 \\ &= a^2(a^{2k+1} - 1) + a^2 - 1 \\ &= a^2 m(a + 1) + (a - 1)(a + 1) \text{ since } a^k - 1 = m(a - 1) \\ &= (a^2 m + a - 1)(a + 1) \end{aligned}$$ **[3]**

This is a multiple of $(a + 1)$ since $am + a - 1$ is an integer, hence $(a + 1)$ is a factor of $a^{2(k+1)+1} - 1$.

$\therefore$ if the conjecture is valid for $n = k$, it is valid for $n = k + 1$. **[1]**

ii No. You must also establish it for a starting value, usually $n = 1$. **[1]**

iii Use a counter example.

For example, if $a = 4$ and $n = 2$ then $a^{2n+1} - 1 = 4^5 - 1 = 1023$ which is not divisible by $a + 1 = 6$.

Other counter examples may be used. **[2]**

iv Proof by induction

Consider $n = 0$.

$a^{2n+1} - 1 = a^1 - 1 = a - 1$

$a - 1 = (a + 1) - 2$ so $a - 1$ will only be a multiple of $(a + 1)$ if 2 is a multiple of $(a + 1)$.

This means $a + 1 = 1$ or $a + 1 = 2 \Rightarrow a = 0$ or $a = 1$

Since $a > 1$, $(a + 1)$ cannot be a factor and so the second conjecture works for $n = 0$. **[2]**

Suppose the conjecture is true for $n = k$.

$(a + 1)$ is not a factor of $a^{2n+1} - 1$.

Then $a^{2k+1} - 1 = m(a + 1) + r$, where r is the remainder and $1 \leq r \leq a$. **[1]**

Consider $n = k + 1$.

$$\begin{aligned} a^{2(k+1)+1} - 1 &= a^{2k+3} - a^2 + a^2 - 1 \\ &= a^2(a^{2k+1} - 1) + a^2 - 1 \\ &= a^2(m(a + 1) + r) + (a - 1)(a + 1) \\ &= a^2 m(a + 1) + a^2 r + (a - 1)(a + 1) \\ &= a^2 m(a + 1) + a^2 r - r + r + (a - 1)(a + 1) \\ &= a^2 m(a + 1) + r(a^2 - 1) + r + (a - 1)(a + 1) \\ &= a^2 m(a + 1) + r(a - 1)(a + 1) + r + (a - 1)(a + 1) \\ &= (a + 1)(a^2 m + r(a - 1) + (a - 1)) + r \end{aligned}$$ **[4]**

Since $1 \leq p \leq a$, $a^{2(k+1)+1} - 1$ will not be divisible by $(a + 1)$ and will actually have the same remainder r as $a^{2k+1} - 1$, the case when $n = k$. **[1]**

$\therefore$ if the conjecture is valid for $n = k$, it is valid for $n = k + 1$.

But, it is valid for $n = 0$ (so valid for $n = 1$, $n = 2$, etc.)

$\therefore$ it is valid for all natural numbers. **[1]**

ALTERNATIVE METHOD

Use the result from part **a**.

$(a - 1)$ will always be a factor of $a^n - 1$. **[1]**

Using a^2 instead of a in this rule gives us the following.

$(a^2 - 1)$ will always be a factor of $(a^2)^n - 1 = a^{2n} - 1$. **[1]**

Thus $a^{2n} - 1 = m(a^2 - 1) = m(a - 1)(a + 1)$ **[1]**

Now consider $a^{2n+1} - 1$ in the conjecture.

$a^{2n+1} - 1 = a^{2n+1} - a + a - 1$ **[1]**

$= a(a^{2n} - 1) + (a - 1)$

$= am(a - 1)(a + 1) + (a - 1)$ **[2]**

Since $a > 1$, $1 \leq (a - 1) < (a + 1)$ and hence $(a + 1)$ cannot be a factor of $a^{2n+1} - 1$.

Alternatively, $a^{2n+1} - 1$ is the sum of a multiple of $(a + 1)$ and a number which is not a multiple of $(a + 1)$, and so cannot be a multiple of $(a + 1)$. **[2]**

v We have unintentionally also shown the remainder will be the same for all values of n as for $n = 0$.

The remainder for $n = 0$ is -2 which we'd normally say is $a + 1 - 2 = a - 1$.

(Remainders are usually between 0 and the divisor.) **[2]**

ALTERNATIVE METHOD (from above)

Since $a^{2n+1} - 1 = am(a - 1)(a + 1) + (a - 1)$, it can be seen that the remainder will always be $r = a - 1$. **[2]**

8 The invalid step is when both sides are divided by $x^2 - xy$

Since $x = y$, $x - y = 0$. **[1]**

$x^2 - xy = x(x - y) = x \times 0 = 0$ and you cannot divide by 0. **[1]**

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